



Listen, Think, and Understand

LTU · an audio foundation model with perception and reasoning

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Audio-language models II

Why this paper matters



The first audio model that can explain itself

5.68M

audio question-answer tuples
across 845K unique clips

4.2M

learnable parameters —
LoRA only

82.9%

instruction-following and
factual correctness rate

+23.6%

average relative gain over
CLAP on eight classification sets

● The claim

Conventional audio models map sound to a fixed label set and cannot elaborate. Large language models can reason about sound in the abstract but cannot hear. Connect an audio spectrogram transformer to LLaMA, train on five million generated questions with a perception-first curriculum, and you get a model that classifies well *and* explains, infers, and refuses.

● The two ingredients the paper says are essential

A dataset containing both closed-ended and open-ended questions, and a perception-to-understanding curriculum. Remove the curriculum and classification collapses by 27 points. Remove the open-ended data and it drops by 13. Both claims are ablated directly.

1

Perception and understanding

The gap, and the question



The gap



Two kinds of model, each missing what the other has

● Audio models can perceive

Since AudioSet, audio tagging mean average precision has risen from 31.4 to 47.3. Perception is strong and getting stronger.

But models trained on discrete label sets — AudioSet's 527 categories — possess very limited reasoning and understanding.

The paper's example: the model may recognise a clock chime six times, but not know that it indicates six o'clock.

● Language models can reason

GPT and LLaMA possess strong audio knowledge and reasoning without any in-domain audio training. Ask ChatGPT what a dog bark sounds like and it will tell you it is short, repetitive and sharp.

But even the most advanced language models cannot perceive general audio events.

They know about sound without being able to hear it.

● The question

Can we build a model that has both audio perception and a reasoning ability?

What humans do that neither side does: listen to the finer details, explain the reason for a prediction, think about what the sound infers, and understand the scene and what action needs to be taken, if any.

2

OpenAQA-5M

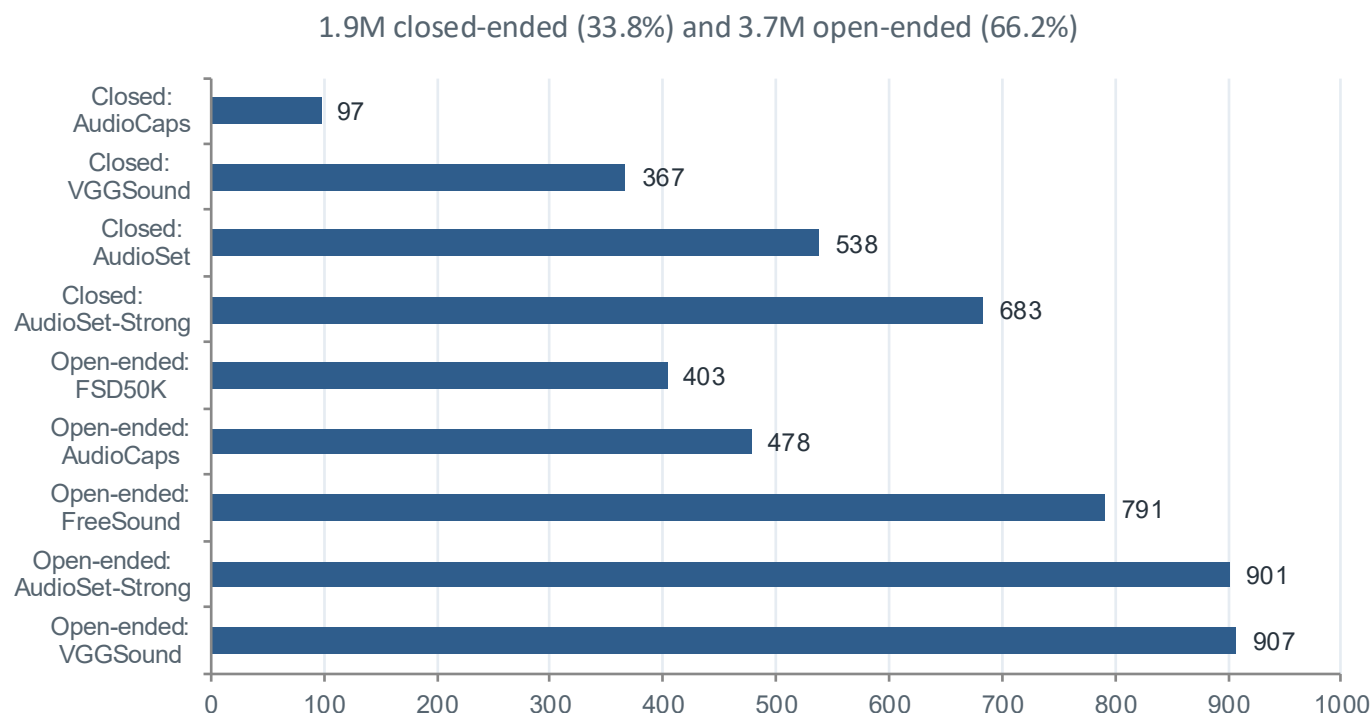
How you make 3.7 million questions about audio



The dataset



Two subsets, eight source corpora, 845,000 clips



- No new audio was collected. Eight existing corpora were relabelled into (audio, question, answer) tuples — AudioSet weak and strong, VGGSound, FSD50K, AudioCaps, FreeSound, Clotho, Sound Bible.
- Only training and validation splits were used; anything marked test or evaluation was excluded.
- The closed-ended half covers four tasks: classification, acoustic features, captioning and temporal analysis.
- Acoustic-feature questions are the largest closed-ended category at 845K pairs — 44% of the closed-ended subset.

Formulating every task as audio question answering is what lets eight heterogeneous corpora become one dataset — Qwen-Audio's interference problem, solved by format rather than by tags.

The four closed-ended tasks



Questions paraphrased by GPT; answers generated by rule

● Classification

'Classify the sound events in the audio clip' and its paraphrases. The answer is the list of sound class names the clip contains.

● Acoustic features

The audio corpora have no such labels, so GPT-3.5 was prompted to 'describe the acoustic characteristic of {class} precisely with a sentence less than 10 words', ten descriptions per class. The answer maps features to classes: 'High-pitched, rapidly changing frequency → Ambulance (siren)'.

● Captioning

Original human captions for AudioCaps and Clotho; WavCaps GPT-assisted captions for FreeSound and Sound Bible.

● Temporal analysis

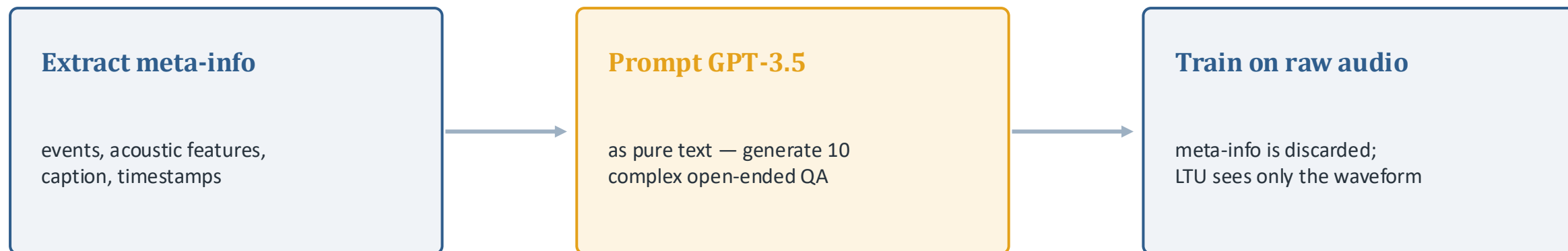
Built from the temporally strong AudioSet labels with a rule-based algorithm: list all timestamps, ask for a specific sound's timestamp, or ask the order of sounds.

Note the design: questions are diverse so the model does not overfit phrasing, answers are rule-generated so they have a fixed, checkable format.

Audio Instruction Generation



The trick: GPT never hears the audio



- The prompt asks for questions requiring step-by-step thinking: which sound is recognised and why, what the temporal relationship implies, the potential scenario, whether the clip is urgent or funny or abnormal and why, what mood it conveys.
- 3.7 million pairs generated — about two orders of magnitude larger than the instruction-tuning sets used for pure language tasks at the time.
- 78% of questions and 95% of answers are unique. Most appear exactly once.
- Crucially, the meta-information is used only in generation and is never input to LTU. The model is forced to learn from raw audio, which contains richer and finer-grained information than the extracted labels.

The 6.5% that cannot be answered



An accident that turned into a feature

● What happened

About 6.5% of the questions GPT-3.5 generated could not be answered from the audio information it was given, so the corresponding answers are of the form 'it cannot be determined from the audio that...'.

These were kept rather than filtered out.

● What they buy

The paper's framing: these questions are equally valuable, because they teach the model to know what it does not know and reduce hallucination.

Remove them from training and the refusal rate on unanswerable questions falls from 69.3% to 51.9%, and closed-ended performance drops slightly too, from 50.3 to 50.0.

● The finding underneath it

Even with almost no unanswerable questions in training, LTU still refuses sometimes — just less often. The authors read this as a behaviour partly learned in the language-model pre-training stage and amplified, not created, by these examples.

A demonstrated example: asked what type of bell is ringing, LTU declines to give a definite answer and lists possibilities instead.

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Architecture and curriculum

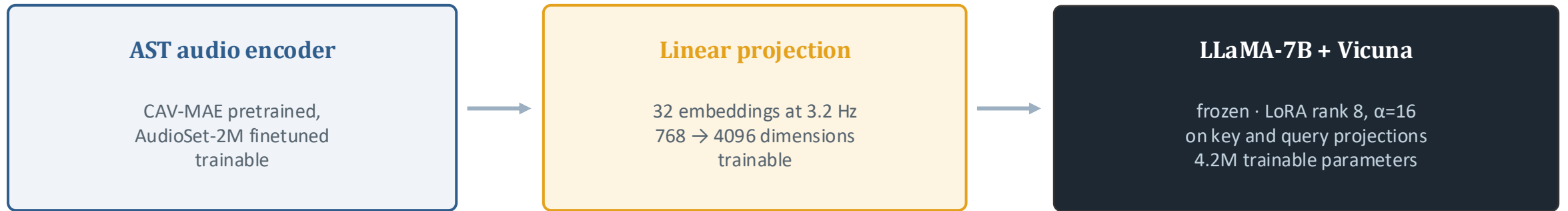
The part that stops hallucination



The architecture



Thirty-two audio tokens into a frozen LLaMA



- Ten seconds of audio becomes a 1024×128 spectrogram, split into 512 patches, encoded to 512 embeddings, then frequency-mean-pooled and temporally downsampled by two to give 32 embeddings — about 3.2 Hz.
- Compare SALMONN at roughly 3 tokens per second and Pengi's fixed 40-embedding prefix. LTU is in SALMONN's range, and the paper flags the low temporal resolution as a limitation.
- LoRA is used rather than full fine-tuning for two stated reasons: mitigating catastrophic forgetting and overfitting, and training efficiency.
- Trained on four RTX A6000 GPUs for about three days — a fraction of Qwen-Audio's compute.

The perception-to-understanding curriculum



Four stages of increasing openness

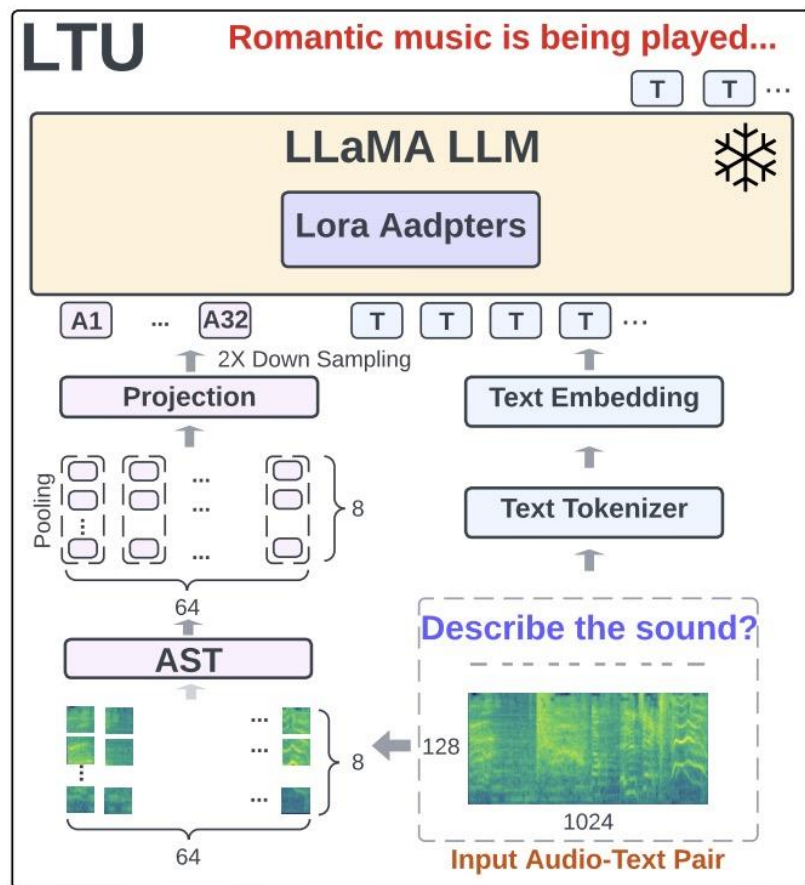
Stage	Trainable	Training task	Samples	Learning rate	Epochs
1	Projection layer only	Classification + acoustic features	1.2M	1e-3	2
2	All (except LLaMA)	Classification + acoustic features	1.2M	1e-4	2
3	All (except LLaMA)	All closed-ended tasks	1.9M	1e-4	1
4	All (except LLaMA)	All closed- and open-ended tasks	5.6M	1e-4	1

● Why the order matters

Open-ended tasks are too difficult for the model at the beginning of training. The model tends to use its language capability to answer the free-form question instead of conditioning on the audio input — that is, to hallucinate.

So closed-ended tasks with objective answers come first, where the model gets a high penalty for wrong predictions. This curriculum teaches the model first to perceive, and then to understand.

The perception-to-understanding curriculum



(Busy Signal Sounds in a Call)

Q: Classify the sound events in the audio clip. [listen]

LTU: Busy signal; Male speech, man speaking; Background noise; Tick; Speech.

Q: What makes you think it is a busy signal? [think]

LTU: The busy signal is characterized by a continuous tone, which is different from the sound of a ringing phone, which is a series of short pulses.

Q: What should the man do? [understand]

LTU: The man should wait for the busy signal to end before trying to call again or try a different phone number if necessary to reach the desired person or organization.



(Owls Night Calls)

Q: What can be inferred from the audio? Think step by step. [listen & think]

LTU: The audio clip contains the sound of wind, insects, and hooting. The wind is blowing and the insects are chirping. The hooting sound is likely from an owl or other bird of prey. The sound of the wind is high-pitched and hissing, while the insects are high-pitched and chirpy. The hooting sound is deep and resonant, suggesting a larger bird of prey or owl. The combination of these sounds suggests a natural outdoor environment, possibly at night. The hooting sound could also suggest a sense of danger or urgency, as owls are known to be predators.

Q: What type of video can this audio be used for dubbing? [understand]

LTU: This audio can be used for dubbing a horror movie.

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Results and ablations

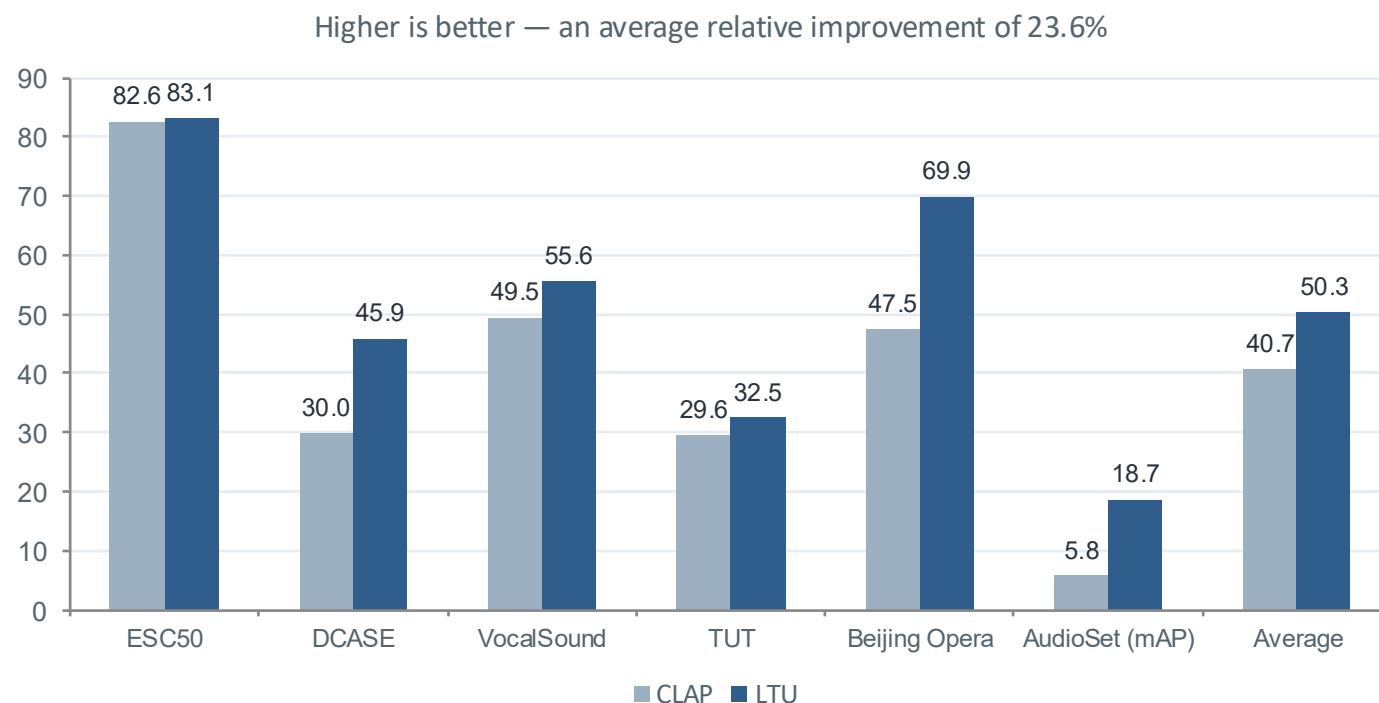
What works, and what breaks when you remove it



Closed-ended performance



Eight classification benchmarks, against CLAP



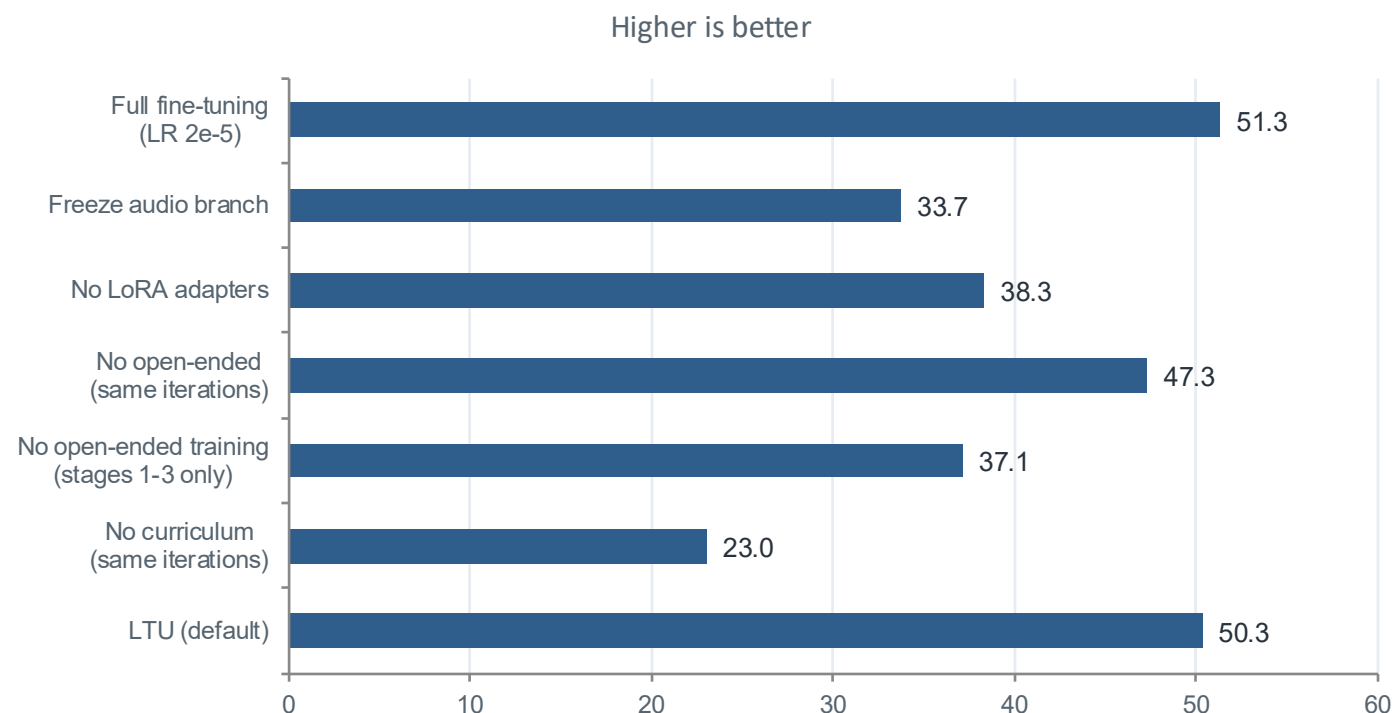
- LTU wins on all eight, and unlike CLAP it needs no predefined label set at inference — it just says what it hears.
- Two honest caveats the paper raises itself: CLAP uses a 44.1 kHz sampling rate against LTU's 16 kHz, which is known to help classification; and mAP underestimates LTU because it only outputs prominent classes rather than a score for all 527.
- Against the best supervised specialist per dataset, LTU is well behind — 50.3 average against 74.3.
- Captioning is competitive: 17.0 SPICE on AudioCaps and 11.9 on Clotho, against specialist bests of 17.7 and 13.5.

The paper's own framing: correct perception is the basis of advanced reasoning. If the model cannot recognise the sounds, its answer to open-ended questions would be nothing but hallucination.

The ablations



Where the performance actually comes from



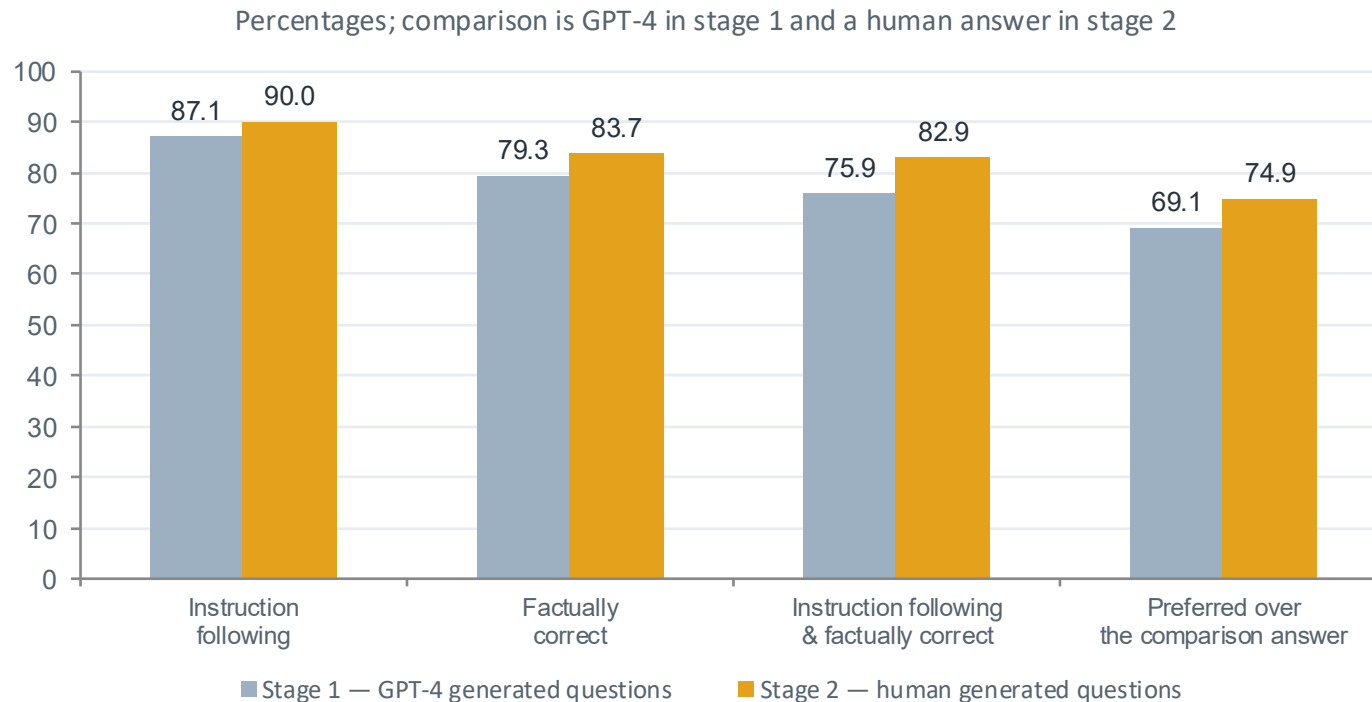
- Removing the curriculum costs 27.3 points, with the same total number of training iterations. This is the single largest effect in the paper.
- Training only on closed-ended tasks costs 13.2 points — so open-ended training improves closed-ended performance. Learning to understand improves the ability to perceive.
- Full fine-tuning scores slightly higher on audio, 51.3, but the language instruction-following rate collapses from 87.6% to 53.5% — catastrophic forgetting, which is why LLaMA is frozen.
- Removing LoRA entirely costs 12.0 points, so the adapters are doing real work, not just providing capacity.

Four controls, four large effects, each testing a stated design decision. This is what the first half of the session was missing.

Evaluating open-ended answers



476 human evaluators, two stages



- 1,352 questionnaires and 476 unique evaluators on Mechanical Turk, independent of the authors.
- Training questions came from GPT-3.5; evaluation questions from GPT-4 and from humans — so the evaluation is not testing memorisation of the generator.
- A separate automatic check with GPT-4 as judge gives an instruction-following rate of 96.9%, against 22.5% for a model trained only on closed-ended tasks.
- Against Pengi, the gap is enormous: Pengi's instruction-following rate on open-ended questions is 7.4%, 3.1% and 28.3% across the three question sources.

The Pengi comparison is the clearest demonstration in either session of what open-ended training buys — and the paper notes Pengi was never designed for this, so the comparison is illustrative rather than fair.

What the reasoning looks like



Five capabilities the paper demonstrates

● Answer follow-up detail questions

Asked how many times a bell rings, LTU says three; asked to describe it, it reports a bright, high-pitched, short ringing repeated in quick succession.

● Explain its predictions

Asked what insect is heard and why: 'a cricket — it is high-pitched and chirpy, which is why it is recognised.' In multi-turn conversation it can explain why a wrong prediction was made.

● Understand scenes and suggest actions

Gunshots and footsteps: LTU infers a war zone or dangerous area, and when asked what to do, says seek shelter and call for help.

● Infer beyond the audio

Owl calls and wind: LTU judges the recording more likely to be evening because owl hooting is prominent at night, and suggests the clip would suit dubbing a horror film.

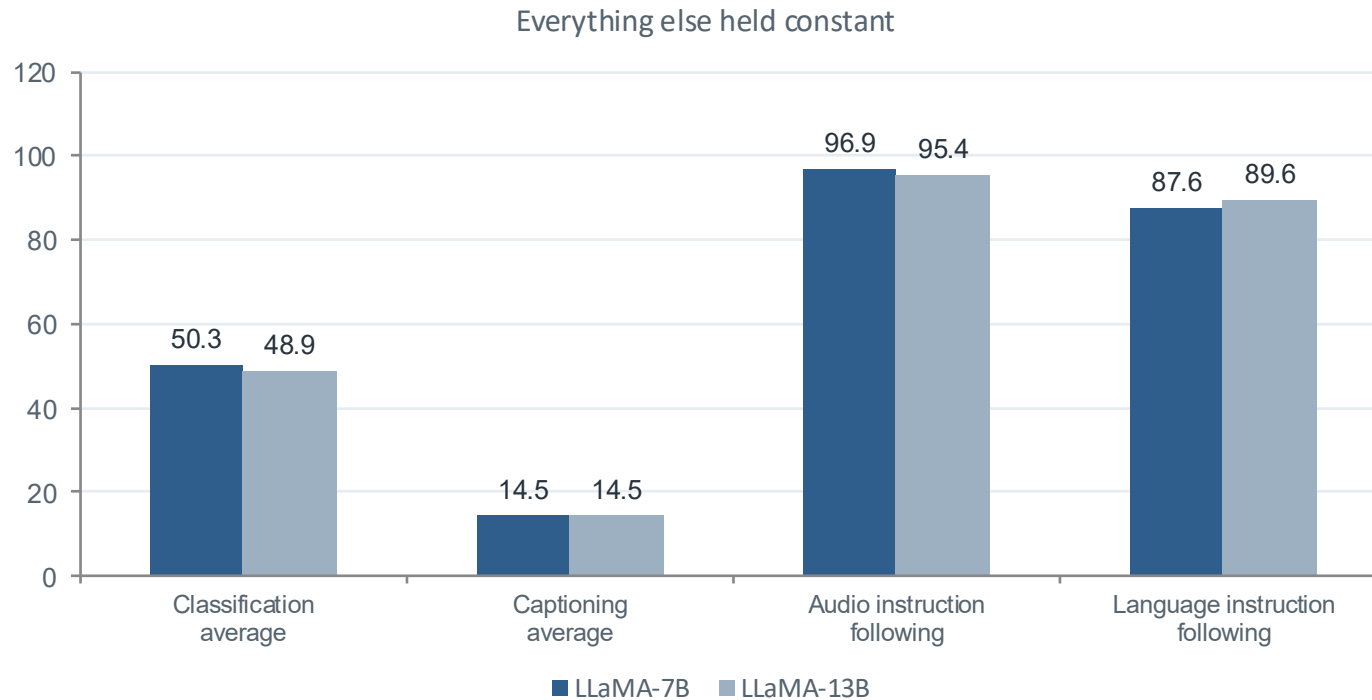
● Know what it does not know

Asked the type of bell, it declines to give a definite answer and lists possibilities — church bell, school bell — rather than guessing.

Does a bigger language model help?



The result that cuts against the scaling narrative



- Classification is slightly worse with the larger model; captioning and instruction following are essentially unchanged.
- But the paper reports that the 13B model gives more rational answers thanks to a better knowledge base, with a worked example about how a power tool would sound in a larger space.
- So the benefit of scale shows up in reasoning quality, which none of the automatic metrics capture.
- The authors list the use of LLaMA-7B — the smallest in the family — as a limitation driven by their computational resources.

A useful corrective to the prompt-tuning session's 'power of scale': here, doubling the language model changes almost nothing that anyone can measure.

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Appraisal and synthesis

The two papers together



Critical appraisal



What the design buys, and what it costs

● Strengths

Audio Instruction Generation is a genuinely reusable idea, and the design detail that matters — metadata is used to generate and then discarded — is what forces the model to learn from audio.

Ablations on every stated design decision, with iteration counts held constant, and effects of 12 to 27 points.

A real human evaluation with 476 independent evaluators, plus a separate automatic check.

Closed-ended results reported first and honestly, with two caveats against the authors' own numbers.

Unanswerable questions kept rather than filtered, with a measured effect on refusal rate.

Code, dataset and pretrained models released; trained on four consumer GPUs in three days.

● Limitations

No speech understanding at all — the paper states plainly it is not a speech recognition model. The successor LTU-AS adds it.

The 'preferred over human answers' figure compares against quick Mechanical Turk responses, and the GPT-4 comparison is against a model that cannot hear.

3.2 Hz audio token rate limits fine-grained temporal reasoning, as the authors note.

Training answers reflect GPT-3.5's beliefs about what sounds are like, uniformly applied per class.

Classification is well behind supervised specialists — 50.3 against 74.3 average.

Hallucination is mitigated, not eliminated, and the refusal-rate measurement is acknowledged as an estimate.

Qwen-Audio and LTU side by side



Perception breadth against understanding depth

	Qwen-Audio (2023)	LTU (2024)
Audio encoder	Whisper-large-v2, 640M, trained	AST via CAV-MAE, trained
Language model	Qwen-7B, fully fine-tuned in stage 2	LLaMA-7B + Vicuna, frozen with LoRA
Trainable parameters	Encoder then full LLM — billions	4.2M LoRA plus the audio branch
Task unification	Six hierarchical tags prepended to the decoder	Everything reformulated as question answering
Training data	30+ tasks, 8 languages, 100k+ hours	5.68M QA pairs from 845K clips, English
Speech ability	State of the art: 2.0 / 4.2 WER	None — explicitly not an ASR model
Open-ended reasoning	Not evaluated	Human-evaluated: 82.9% correct and on-instruction
Ablation of headline claim	None	Curriculum removed: -27.3 points
Compute	Industrial scale, 500k pre-training steps	4 RTX A6000s, 3 days
What it demonstrates	One model can perceive everything	A small model can explain what it hears

The highlighted row is the methodological difference that matters most for how much each paper can claim.

Take-aways



Four things to carry out of this session

1 Generate the data you need from metadata

GPT never heard the audio; it wrote questions from labels, features and timestamps, and the metadata was then discarded. This is now the standard recipe for audio instruction data.

2 Order the curriculum by openness

Closed-ended tasks first, where a wrong answer is punished hard, then open-ended. Removing that ordering costs 27 points with the same compute.

3 Understanding improves perception

Training on open-ended reasoning made the model better at plain classification. The direction of transfer is not the obvious one.

4 Teach the model to refuse

6.5% unanswerable questions raise the refusal rate from 51.9% to 69.3%. Keeping what a naive pipeline would filter out was the intervention.

Questions for discussion



1

LTU's acoustic-feature vocabulary comes from GPT-3.5 describing what each class sounds like, applied uniformly to every clip of that class. What is the model actually learning, and how would you test whether those descriptions correspond to anything in the signal?

2

Open-ended training improved closed-ended classification by 13 points. What is the mechanism, and would you expect the same direction of transfer in vision or in speech emotion?

3

The evaluation compares LTU against GPT-4 answering from metadata and against quick crowd-worker answers. What would a defensible baseline for open-ended audio reasoning look like?

4

Doubling the language model changed almost nothing measurable, but the authors report better reasoning. Is that a metric failure or an anecdote — and what would settle it?

5

Qwen-Audio perceives across four audio types and never reasons; LTU reasons and cannot hear speech. Both are 2023-24 systems with the same architecture. Why did nobody build both at once, and what would it take?